

Collaborator or Assistant? How AI Coding Agents Partition Work Across PR Lifecycles

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$V = 0.50$

What does Cramer's V measure?

Tool identity predicts PR collaboration scenario: who starts work, who authorizes merge, and where governance remains human.

29,585

lifecycle traces

PRs

5

tools

compared

6

paths

scenario table

Collaborator or Assistant? Chung & Hassan | University of Toronto

A spectrum with partial decoupling

INITIATION

Tools separate sharply

Collaborator tools initiate most PRs.

Assistant tools mostly enter human-started PRs.

AUTHORIZATION

Merge stays human

Agent-classified approvers are rare in this snapshot.

Governance is observable but not reducible to operation.

INTERPRETATION

Related, not locked

Operational work and merge governance can diverge.

That divergence is the measurement target.



Today's agenda

1

GAP

Prior work vs lifecycle reconstruction

2

METHOD

Pipeline, trace vocabulary, observability bounds

3

SPECTRUM

Taxonomy, $V = 0.50$, decoupling

4

WORKFLOWS

State machines: Copilot vs OpenAI

5

IMPLICATIONS

Contributions and future layers

RQ1-RQ3

research questions

partitioning, workflows,
automation boundaries



What prior work measures, and what we add

Outcome metrics

merge rates and acceptance

not who did what, when

Single-phase studies

suggestions and comments

not full lifecycle

Governance frameworks

oversight design

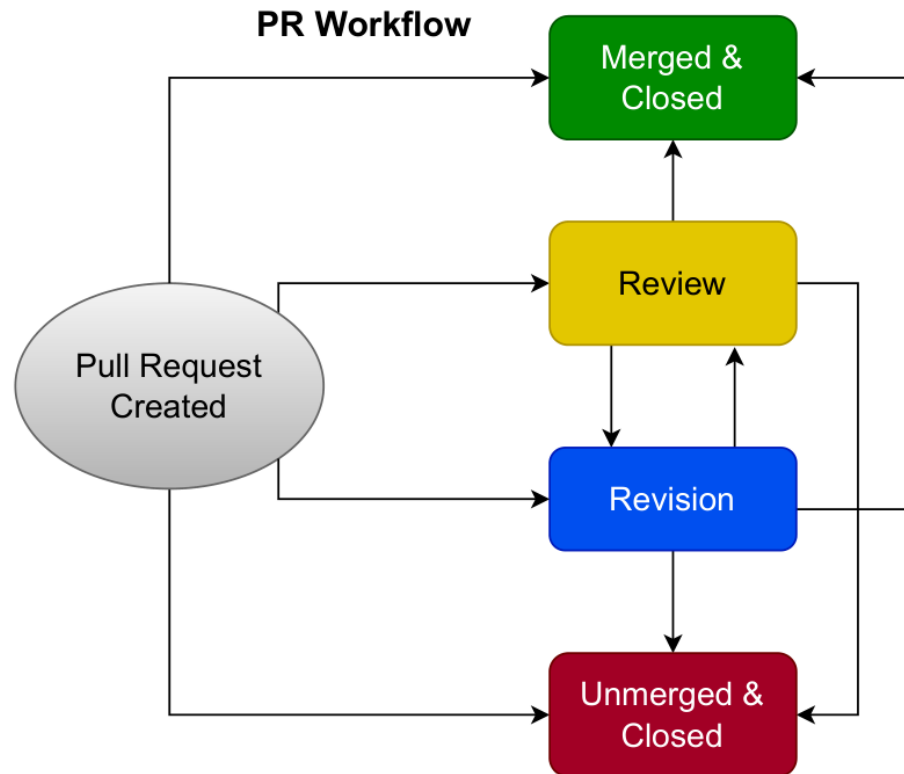
rarely at scale in traces

OUR ADD

29,585 PR lifecycles
initiator to approver
state machines
replication package



Lifecycle model + pipeline

**29,585 PRs**

sample

5

tools

state machines

output

29,585 PRs | 5 tools | AIDev (Li et al., 2025)



What the trace lets us reconstruct

AGENTS

who acts

initiator and approver on each PR

ROLES

what work

operational work vs merge governance

DEPENDENCIES

how flow moves

review routing and revision loops

PROTOCOLS

what rules

branch protection, merge rules, automation bots



What GitHub traces can observe

CAN OBSERVE

committed, review_requested, reviewed,
merged/closed events

actor classification: Bot flag + pattern heuristic

phase transitions, terminal outcomes, median
hours

CANNOT INFER FROM TRACES ALONE

developer intent

off-platform decisions

decision vs executor on auto-merge

0.36% FP

actor heuristic
validation

0 FN

false negatives



Illustrative composite: Collaborator PR (S1)

STEP 1

Agent commits

agent-authored changes
appear in the PR history

STEP 2

PR opened

the lifecycle trace
begins as an observable
review object

STEP 3

Human review

review routing makes
governance work visible

STEP 4

Human merge

authorization remains
with a human approver

Illustrative composite - aggregate Copilot-style S1 pattern synthesized from scenario definitions; not a single quoted repository PR.



Initiator x Approver: six scenarios

AGENT-INITIATED

S1 / S2 / S3

human-approved
agent-approved
not merged

HUMAN-INITIATED

S4 / S5 / S6

human-approved
agent-approved
not merged

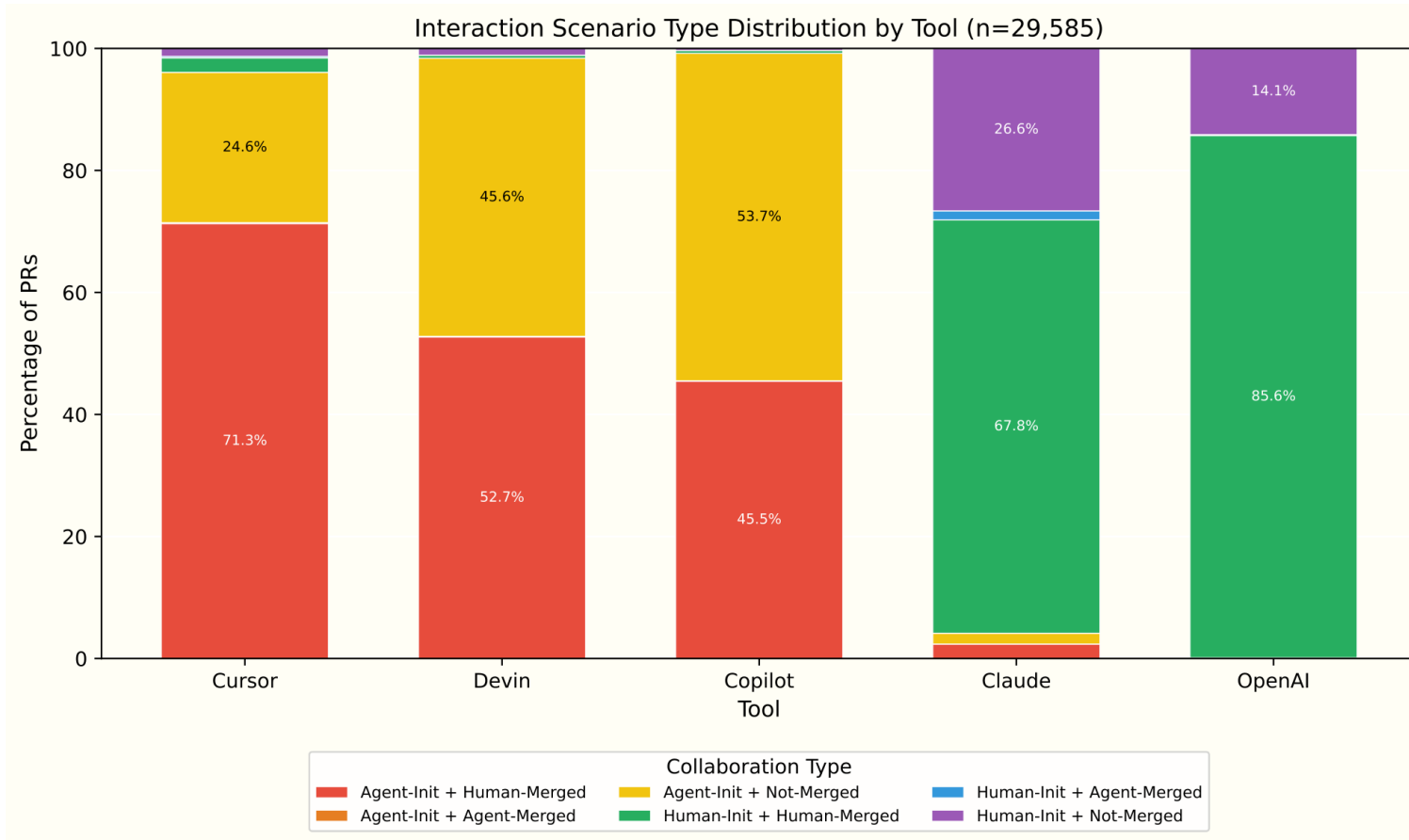
TOOL SPLIT

$\geq 90\%$

Collaborators cluster in S1-S3
Assistants cluster in S4-S6



The Collaborator-Assistant spectrum



Cramer's $V = 0.50$; chi-square rejects independence at $p < 0.001$.

0.50

V

≥ 0.25 large

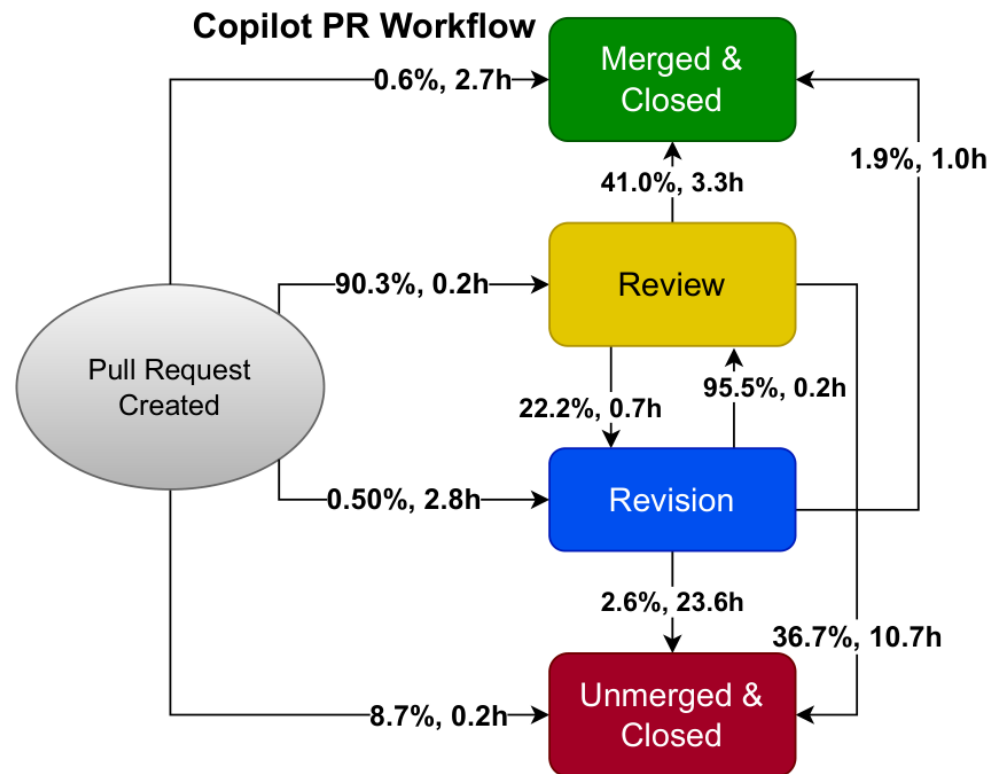
threshold

review vs direct

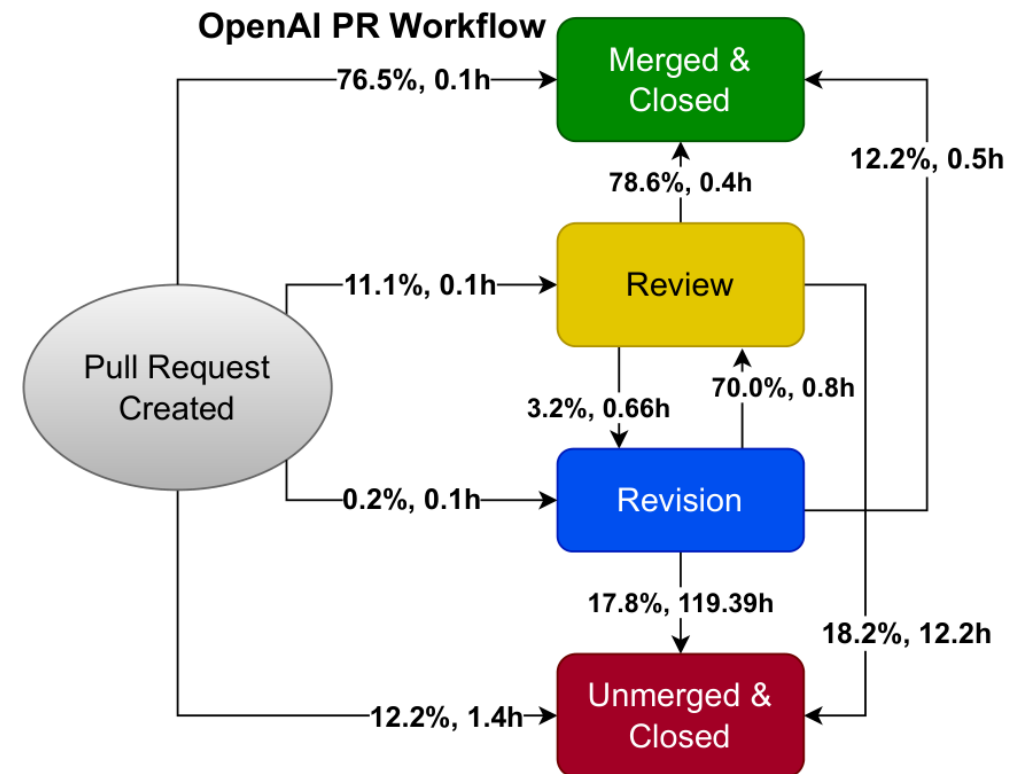
contrast



Copilot vs OpenAI: workflow state machines



Review-heavy Collaborator



Direct-resolution Assistant



Agency and governance decouple

$\geq 96\%$

Collaborator tools
agent-initiated PRs

$< 0.05\%$

agent-approved (S2)
of all 29,585 PRs

14 PRs

S2 cases
rare in this snapshot

Junior-teammate pattern: agents operate; humans authorize merge.

Endorsement is measurable here; accountability is the layer the paper does not measure.



Policy and snapshot bounds

BRANCH PROTECTION

shapes routing

direct resolution is partly repository governance

WITHIN-REPO CONTROL

96.8 pp

initiation gap persists; review routing partly reverses

SNAPSHOT

94%

of PRs from 2025 Q2-Q3; not a trend claim



What traces cannot settle

54 PRs

agent-classified approvers

0.24% of merged PRs

automation bot

merge actor

often executor, not coding agent

decision-maker

taxonomy boundary

not always visible in logs

Logs record executor, not always decision-maker.

Different denominators: 14 of all, 54 of merged.



Contributions, scope, and limitations

CONTRIBUTIONS

what we ship

Initiator x Approver taxonomy
five workflow state machines
public replication package

SCOPE

what we cover

AIDev cross-sectional snapshot
tool-adopting repositories
Q2-Q3 2025

LIMITS

what we honestly admit

observability ceiling
policy confound
Claude small-n; 0.36% FP rate



Implications and future research

TEAMS

lifecycle-based review

route agent-opened PRs differently
from human-opened PRs

RESEARCHERS

workflow-situated

complements task-realistic
benchmarks
asks how code moves through
governance

TOOL BUILDERS

decision provenance

expose who decided, who
executed, under which rule

Method portable to any human-AI workflow with logged events. PRs are one operationalization.



Q & A

Lifecycle traces for governance in AI-assisted pull requests.



REPLICATION PACKAGE

doi.org/10.6084/m9.figshare.31343038

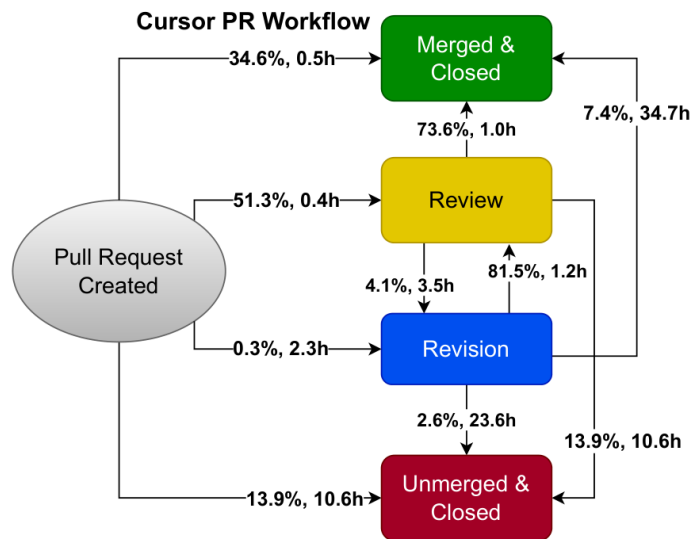
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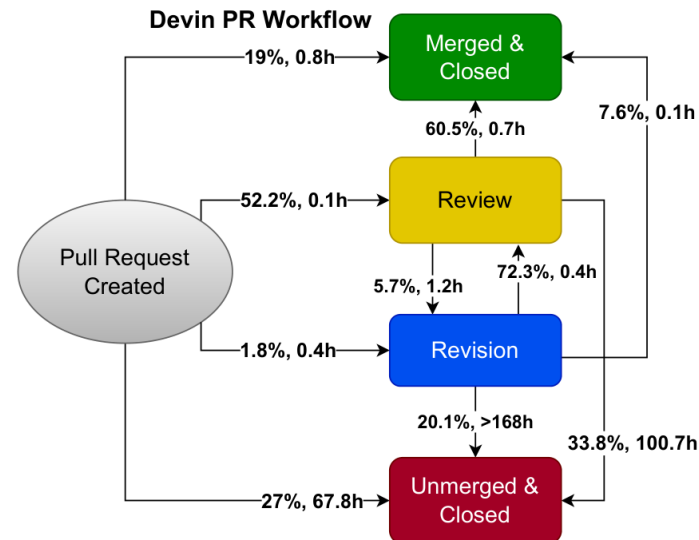
Thank you.



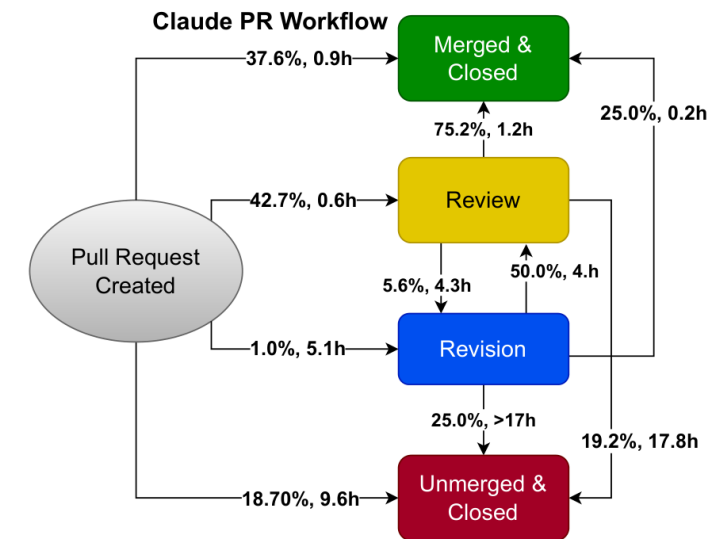
Backup B1: Cursor, Devin, Claude state machines



Cursor



Devin



Claude

